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To the Editor-in-Chief, *Artificial Intelligence Review*

Re: Submission of a review article — “When Models Know They Are Being Tested: A Scoping Review of Evaluation Awareness, Sandbagging and Related Model-Originated Threats to the Validity of Large Language Model Evaluation”

Dear Editor,

We submit the above manuscript for consideration as a review article in *Artificial Intelligence Review*.

Decisions about deploying and governing large language models increasingly rest on evaluation results. A rapidly growing literature, most of it from 2025–2026, reports that models can recognise when they are being evaluated and may behave differently as a result. They may underperform on capability tests, appear safer on alignment tests, or mislead automated graders. These are threats that originate in the evaluated model itself rather than in benchmark design. To our knowledge they have not been synthesised as a single validity problem: existing reviews address AI deception broadly, benchmark construct validity, reward hacking, or evaluation–safety failures in general, and the one review of the AI “observer effect” predates the 2025–2026 measurement literature.

The manuscript contributes:

1. A tiered evidence map of 167 core and 148 adjacent studies, identified through Scopus, arXiv, Semantic Scholar and logged web searches (5649 database and 418 web records), with the protocol and amendments, all queries, raw API responses, screening decisions and counts released as reproducible scripts and data. The full texts of 37 anchor studies were read, every quantitative claim about them was checked, and their reporting was audited against a twelve-item checklist with independent double coding.
2. A four-layer framework (context signal, awareness, behavioural divergence, validity consequence), formalised as a decomposition of evaluation error that states what each study design can and cannot identify.
3. A comparison of measurement methods with their validity evidence and documented failure modes, and a typed analysis of seven apparent contradictions.
4. An evidence-graded research agenda and a twelve-item reporting checklist for evaluation papers and system cards.

We believe the work suits the journal’s readership, which spans AI methods and their critical evaluation, because it addresses how the community knows what its models can do and how they behave.

The manuscript is transparent about its limitations. Screening and charting were performed by an AI system against written criteria under the authors’ direction, without a second human reviewer; we report consistency statistics for these AI-assisted processes (not human inter-rater reliability) and list a human audit as the main outstanding check. Full texts were read for the anchor set only. As set out in the AI-use statement, an AI system was used extensively to run and log searches, apply scripted screening and charting rules, generate figures, and draft text. The authors directed the work and take full responsibility for it.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. Both authors have approved the manuscript and agree with its submission. The authors declare no

competing interests.

Thank you for your consideration.

Yours sincerely,

Divya Meena S (corresponding author), on behalf of both authors

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